

Newton Method H2

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$$f(x, y) = x^2 + e^y - 4y, \quad x^0 = (3, 0)$$

$$\nabla f(x, y) = (2x, e^y - 4)$$

$$H = \begin{bmatrix} 2 & 0 \\ 0 & e^y \end{bmatrix}$$

$$\nabla f(x, y) = 0 \Rightarrow \begin{cases} 2x = 0 \\ e^y - 4 = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \ln 4 = 1.3863 \end{cases}$$

$$x^* = (0, 1.3863)$$

$$\nabla f(x^*) = 0^T + 1 - 4 \cdot 1.3863 = 4 - 5.5452 = -1.5452$$

$$H(x^*) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \Rightarrow \det = 8 \Rightarrow 8 > 0, 2 > 0 \Rightarrow H > 0 \Rightarrow \text{PD, strict min.}$$

$$x^0 = 0, \quad x^0 = (3, 0)$$

$$f(x^0) = 3^2 + e^0 - 4 \cdot 0 = 9 + 1 - 0 = 10$$

$$\nabla f(x^0) = (2 \cdot 3, e^0 - 4) = (6, -3)$$

$$H(x^0) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$H \cdot p = -\nabla f \Rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix} \Rightarrow \begin{matrix} p_1 = -3 \\ p_2 = 3 \end{matrix}$$

$$\langle \nabla f, p \rangle = 6 \cdot (-3) + (-3) \cdot 3 = -18 - 9 = -27 - p \text{ is a descent direction}$$

$$\|y^0 - y^*\| = |0 - 1.3863| = 1.3863$$

$$x^1 = x^0 + p^0 = (3-3, 0+3) = (0, 3)$$

$$k=1, x^1(0, 3)$$

$$f(x^1) = 0^2 + \frac{1}{4} \cdot 3^2 = 0 + 2.0855 = 2.0855$$

$$\nabla f(x^1) = (2 \cdot 0, 2 \cdot 0.0855) = (0, 0.171)$$

$$H(x^1) = \begin{bmatrix} 2 & 0 \\ 0 & 0.171 \end{bmatrix}$$

$$Hp = -\nabla f \Rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 0.171 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.171 \end{bmatrix} \Rightarrow \begin{matrix} p_1 = 0 \\ p_2 = -0.80085 \end{matrix}$$

$$\min. \langle \nabla f, p \rangle = 0 \cdot 0 + 0.171 \cdot (-0.80085) = -0.1369$$

$$\langle \nabla f, p \rangle = 0 \cdot 0 + 0.171 \cdot (-0.80085) = -0.1369$$

$$\|y^1 - y^*\| = |3 - 1.3863| = 1.6137$$

$$x^2 = (0, 3 - 0.80085) = (0, 2.19915)$$

$$k=2, x^2(0, 2.19915)$$

$$f(x^2) = 0 + 9.0173 - 4 \cdot 2.19915 = 9.0173 - 8.7966 = 0.2207$$

$$\nabla f(x^2) = (0, 9.0173 - 4) = (0, 5.0173)$$

$$H(x^2) = \begin{bmatrix} 2 & 0 \\ 0 & 5.0173 \end{bmatrix}$$

$$\nabla f = -\nabla f = \begin{bmatrix} 2 & 0 \\ 0 & 0.0173 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5.0173 \end{bmatrix} \quad \begin{matrix} p_1 = 0 \\ p_2 = -0.5564 \end{matrix}$$

$$\langle \nabla f, p \rangle = -\cancel{(5.0173)^2} \cdot 0 \cdot 0 + 5.0173 \cdot (-0.5564) = -2.7919$$

$$|y^k - y^*| = |2.99915 - 1.38631| = 0.8129.$$

(a) **Monotonicity.** The objective decreases along the computed iterates: $10.0000 > 8.0855 > 0.2207$. However, the error $|y - y^*|$ initially increases from 1.3863 to 1.6137, then decreases to 0.8129.

(b) Armijo condition. At $k = 0$ and $\alpha = 1$, $f(x^0 + p^0) = 8.0855 \leq 10 + 10^{-4}(-27) = 9.9973$. Therefore, damped Newton accepts the full step.

(c) Error ratio. For $k = 1$, $|y^2 - y^*| / |y^1 - y^*|^2 \approx 0.3122$. The theoretical asymptotic ratio is $1/2$; these iterates are not yet sufficiently close to the minimizer. One measured ratio alone does not establish quadratic convergence.

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